



Getting Started with Geographic Information Systems (GIS)

Intro to GIS + Map Creation Basics

WORKSHOP

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What We'll Cover Today

Part 1 — This Session

- What GIS is and why it matters
- Basic concepts of spatial data
- How to create a simple map

Part 2 — Hands-On Session

- Building a map step-by-step
- Adding and styling data layers
- Exploring data and creating a final map product

What is GIS?

Geographic Information Systems (GIS) is a technology for capturing, storing, analyzing, and visualizing geographic data. In simple terms: **GIS connects data + location** to help us understand the world.

Capture

Collect geographic data from the real world

Store

Manage and organize spatial information

Analyze

Discover patterns and relationships

Visualize

Display insights on interactive maps



What Makes GIS Powerful?

GIS helps answer critical questions — and is used across many fields.



Where are things?

Locate resources, hazards, or services precisely.



What patterns exist?

Reveal trends invisible in spreadsheets.



How do places relate?

Understand spatial relationships between locations.



What has changed?

Track changes over time across a landscape.



Used in: Public health · Disaster response · Transportation · Equity analysis · Campus planning

Core Idea: Data + Location

Every GIS dataset has two essential parts working together.

Spatial Data — Where

- **Points** — trees, buildings, schools
- **Lines** — roads, rivers, pipelines
- **Polygons** — boundaries, campuses, zones

Attribute Data — What

- **Name** — label or identifier
- **Type** — category or classification
- **Population** — numeric values
- **Status** — condition or state
- **Any description** — additional context

Key GIS Components



Basemap

The background reference layer — streets, terrain, or satellite imagery.



Data Layers

Your custom geographic information overlaid on the basemap.



Symbology

Colors, shapes, and sizes that represent different data values.



Pop-ups

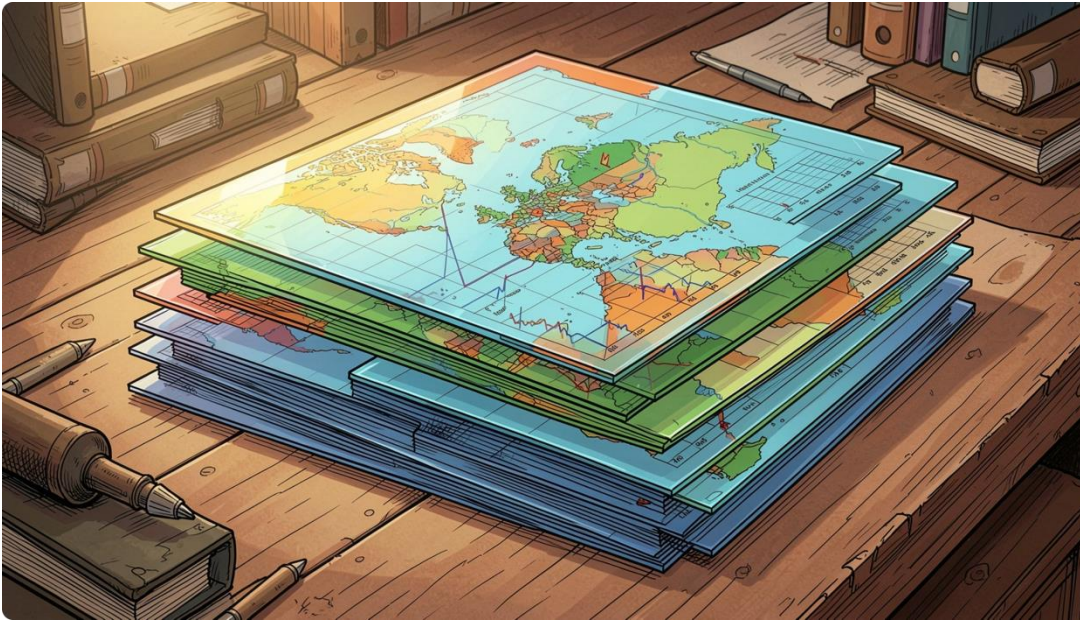
Information windows that appear when you click a feature.



Legend

A guide explaining what each symbol and color means.

What is a Map in GIS?



A GIS map is **more than a static image**. It's a dynamic collection of **layers** — each holding a different dataset that can be turned on or off, styled, and analyzed.

- i Think of it like a stack of transparent sheets. Each sheet represents a different dataset — roads, buildings, parks — layered together to tell a complete story.

How to Make a Map: Basic Workflow

01

Define Your Question

What are you trying to show? Who is the audience?

02

Find or Load Your Data

CSV files, GIS servers, open data portals, or institutional datasets.

03

Add a Basemap

Choose a background reference map.

04

Add Data Layers

Load your spatial data as points, lines, or polygons.

05

Style Your Layers

Apply colors, symbols, and sizes to represent your data.

06

Configure Pop-ups

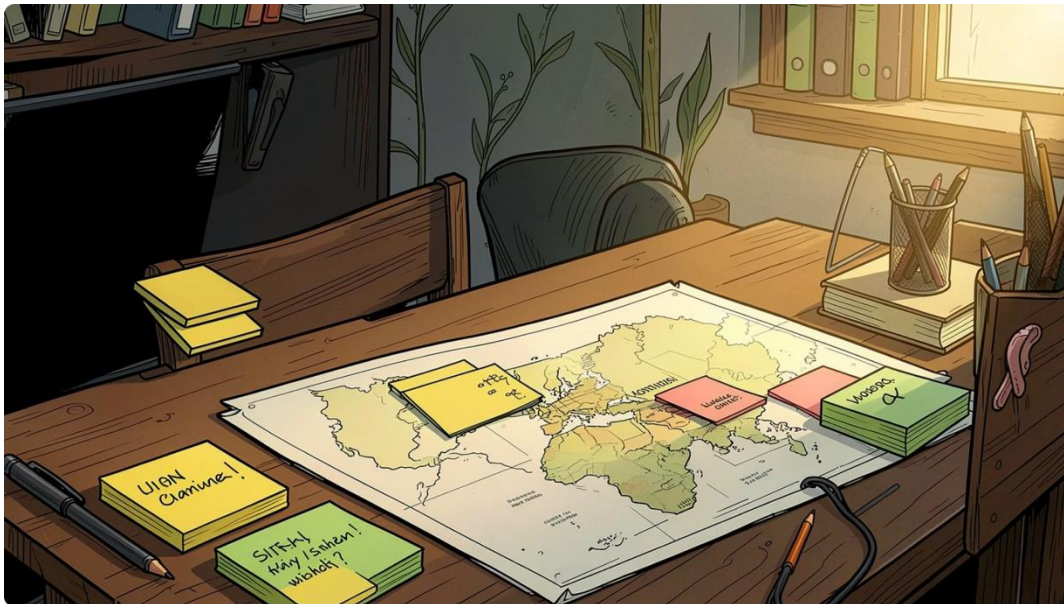
Set up information windows for map features.

07

Review and Interpret

Check your map and draw conclusions from what you see.

Step 1: Define Your Purpose



Before building any map, start with a clear purpose.
Ask yourself:

What am I trying to show?

Define the core message or insight.

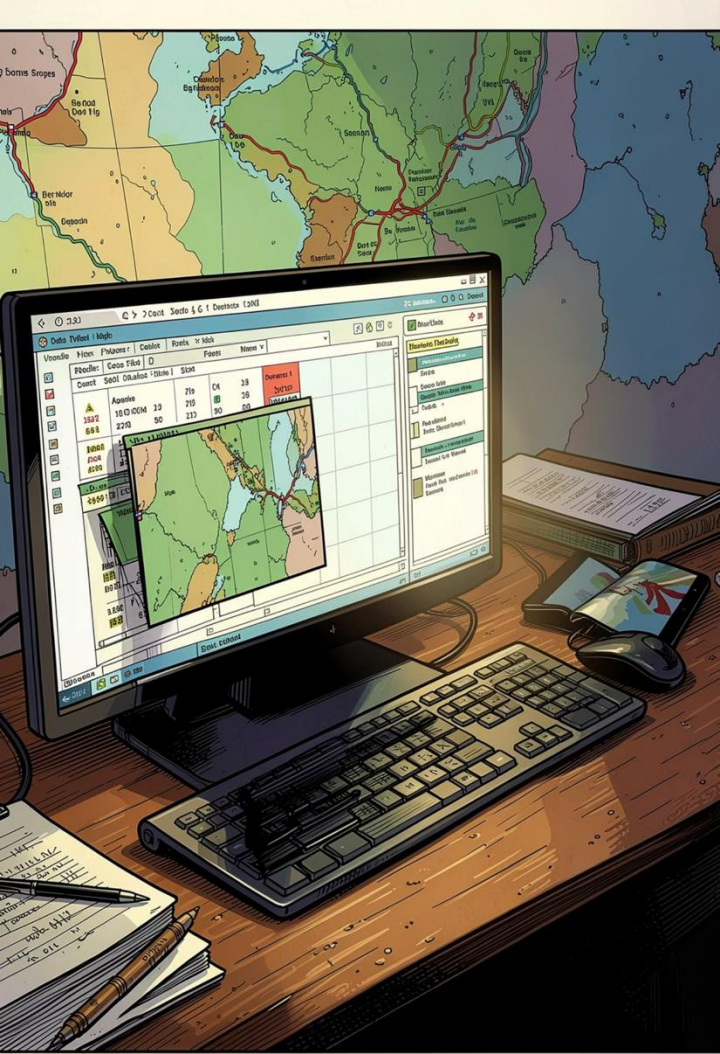
Who is the audience?

Tailor complexity and design to the viewer.

What decision will this support?

Maps should drive action or understanding.

☐ **Example questions:** Where are campus resources? How are services distributed? Where are environmental risks highest?



Step 2: Add Your Data

Data is the foundation of every GIS map. You can bring data in from many sources.



Spreadsheets (CSV)

Tables with addresses or coordinates



GIS Servers

Live data feeds from institutional systems



Open Data Portals

Public datasets from governments and organizations

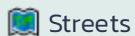


Institutional Datasets

Campus, city, or agency-specific data

Step 3: Choose a Basemap

A basemap is the foundation of your map — it provides geographic context so your data has somewhere to live.



Streets

Roads, labels, and boundaries — ideal for navigation and urban data.



Satellite

Real-world imagery — great for land use, environment, and context.



Terrain

Elevation and topography — useful for natural features and landscapes.

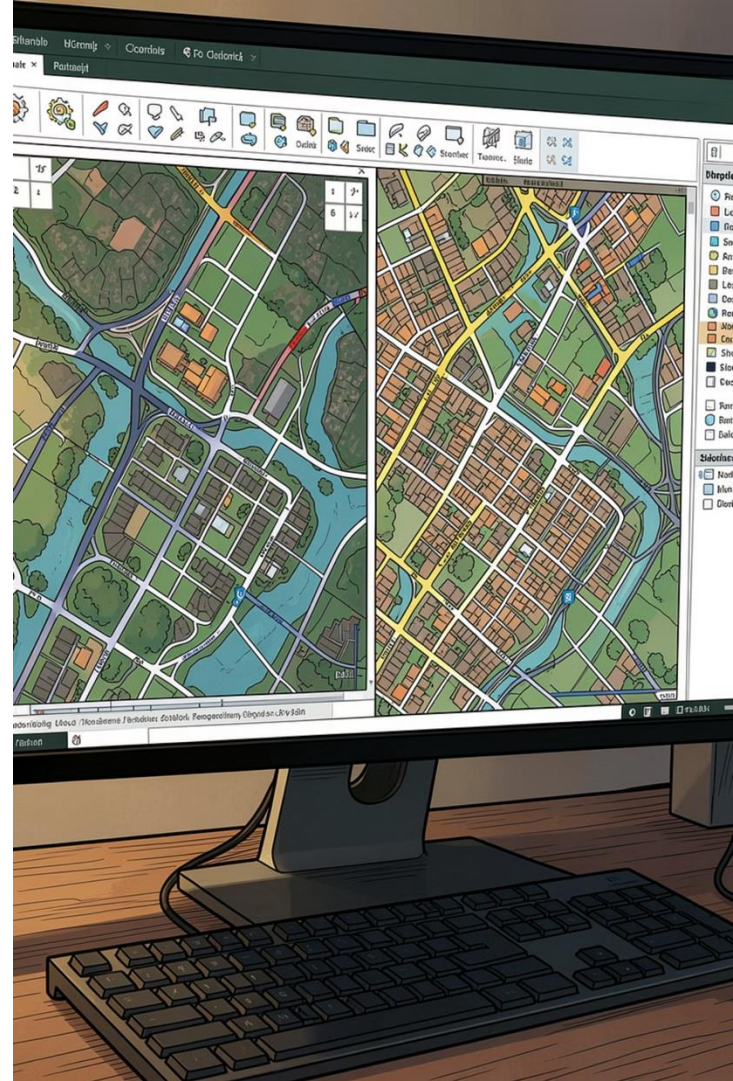


Canvas

Minimal light or dark backgrounds — lets your data take center stage.

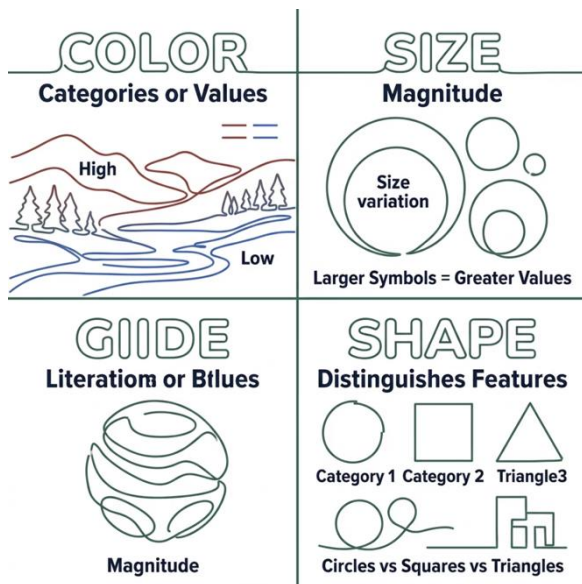


Why it matters: The right basemap helps your audience immediately understand where they are and what your data means in context.



Step 4: Style Your Data

Styling turns raw data into a visual story. Thoughtful design choices help your audience understand patterns at a glance.

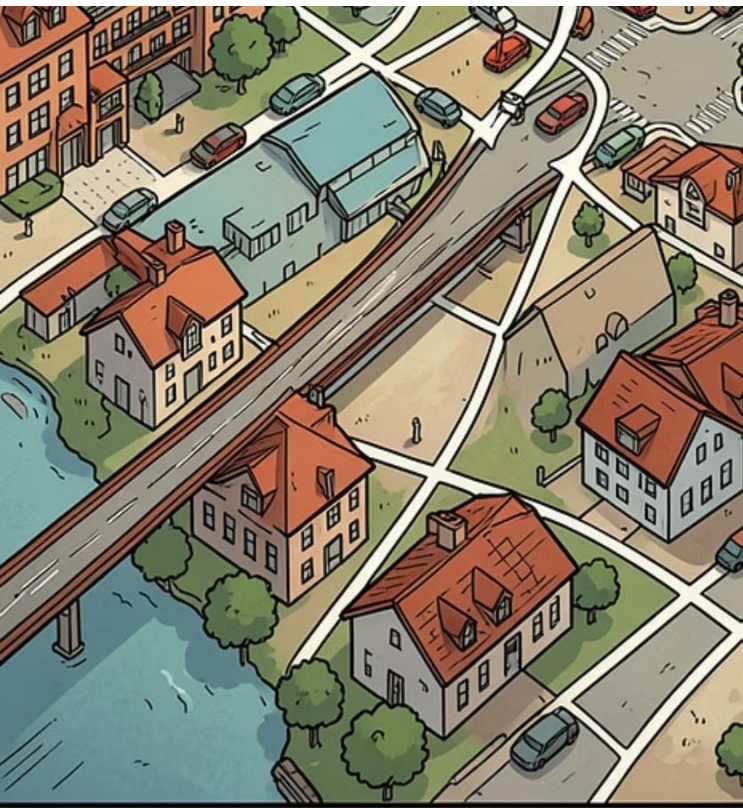


Design with Purpose

Every styling choice should serve your message:

- **Color** — distinguish categories or show value ranges
- **Size** — communicate magnitude or importance
- **Shape** — differentiate types of features

❏ Goal: Make patterns easy to understand without reading a legend.



Step 5: Add Meaning with Pop-ups & Labels

Maps become truly useful when people can explore them. Pop-ups and labels turn a static image into an interactive experience.

Pop-ups

Users click a feature to reveal details — names, values, descriptions, or links. Pop-ups make your map exploratory and informative.

Labels

Key information displayed directly on the map — city names, site identifiers, or important features visible at all times.

What You've Built So Far

You now have the core building blocks of a functional GIS map. Here's a quick recap of your progress:

01

Basemap Selected

A geographic foundation that gives your data context and orientation.

03

Basic Styling Applied

Colors, sizes, or shapes that communicate meaning visually.

02

Data Layers Added

One or more layers representing real-world features or phenomena.

04

Spatial Understanding

A foundational grasp of how geographic data is structured and displayed.



Now It's Your Turn

Time to move from theory to practice. In this hands-on session, you'll build a real map from scratch.

1

Open Your Tool

Launch the GIS mapping platform.

2

Load Real Data

Import a dataset relevant to your campus or community.

3

Build & Style

Apply basemaps, layers, and styling step-by-step.

4

Explore & Ask

Use spatial thinking to answer real questions with your map.



Your facilitator **Daniel** will now guide you through the practical exercise. Don't hesitate to ask questions!